Computational Mathematics

Sheet II.2 — HT25

1. The algorithm for multiplication studied in school, known as long multiplication, consists of multiplying each digit in the first number by each digit in the second, and adding the results. What is the time complexity of long multiplication applied to two inputs, each with n digits?

2. Let $f \in C^\infty([a,b],\mathbb{R})$. Assume that Newton's method from a fixed initial guess $x_0 \in [a,b]$ gives a sequence of iterates which remain in $[a,b]$ and converges to $\alpha \in [a,b]$.
   (a) Show that Newton's method is at least quadratically convergent to α if $f'(\alpha) \neq 0$.
   (b) Show that Newton's method is linearly convergent to α if $f'(\alpha) = 0, f''(\alpha) \neq 0$. What is the rate of convergence in this case?
   (c) How does the rate of convergence vary if $f'(\alpha) = 0, f''(\alpha) = 0, f'''(\alpha) \neq 0$?
   (d) What is the order of convergence of Newton's method if $f'(\alpha) \neq 0, f''(\alpha) = 0$?

3. Apply Newton's method with $x_0 = 1/2$ to find the root $\alpha = 0$ of the following functions:
   (a) $f(x) = \sin x \sinh x$
   (b) $f(x) = x - \sin x$
   (c) $f(x) = x \cos x$

   Terminate the iteration when $|f(x)| < 10^{-8}$. In each case, relate the order of convergence observed to the previous question.

4. (a) Implement the multidimensional Newton iteration to compute a root of $F : \mathbb{R}^N \to \mathbb{R}^N$. Your code should be called like `newton(F, J, x0, tol)`, where `F` is the residual, `J` is the Jacobian, `x0` is the initial guess, and `tol` is the convergence tolerance.

   [*Hint: use* `numpy.linalg.solve` *to solve the arising linear system.*]

   (b) Consider the system of equations

$$x^2 - y = 1$$
$$x - y^2 = -1$$

   Using your code from part (a), find at least two solutions of this system. How many solutions can you find?

   [*Hint: you could use* `sympy` *for calculating the entries of the Jacobian matrix.*]

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2. Let $f \in C^\infty([a,b],\mathbb{R})$. Assume that Newton's method from a fixed initial guess $x_0 \in [a,b]$ gives a sequence of iterates which remain in $[a,b]$ and converges to $\alpha \in [a,b]$.

(a) Show that Newton's method is at least quadratically convergent to α if $f'(\alpha) \neq 0$.

(b) Show that Newton's method is linearly convergent to α if $f'(\alpha) = 0$, $f''(\alpha) \neq 0$. What is the rate of convergence in this case?

(c) How does the rate of convergence vary if $f'(\alpha) = 0$, $f''(\alpha) = 0$, $f'''(\alpha) \neq 0$?

(d) What is the order of convergence of Newton's method if $f'(\alpha) \neq 0$, $f''(\alpha) = 0$?

a)

Let $g(x) = x - \dfrac{f(x)}{f'(x)}$

Taylor expanding around the fixed point $d \in [a,b]$

$$g(x_i) = g(d) + (x_i - d)g'(d) + \frac{1}{2}(x-d)^2 g''(\xi) \qquad \xi \in (x_i, x^*)$$

$$\Rightarrow |x_{i+1} - d| = \left|(x_i - d)g'(d) + \frac{1}{2}(x_i - d)g''(\xi)\right|$$

$$\leq |x_i - d|\, g'(d) + \frac{1}{2}|x_i - d|^2 \max_{\xi \in (x_i, d)} |g''(\xi)|$$

$$g'(x) = 1 - \frac{(f'(x))^2 - f(x)f''(x)}{(f'(x))^2}$$

$$= \frac{f(x)f''(x)}{f'(x)^2}$$

$\Rightarrow$ if $f'(d) \neq 0 \Rightarrow g'(d) = 0$

as $f(d) = 0$

$$\Rightarrow |x_{i+1} - d| \leq \frac{1}{2}|x - d|^2 \max |g''(s)|$$

$$\Rightarrow \frac{|x_{i+1} - d|}{|x_i - d|^2} \to C$$ so we have at least quadratic convergence.

Say instead that $f'(d) = 0$

then $g'(d) = \lim_{x \to d} \frac{f(x)f''(x)}{(f'(x))^2}$ by L'hopital

$$\lim_{x \to d} \frac{f'(x)f''(x) + f(x)f'''(x)}{2f'(x)f''(x)}$$

$$= \frac{1}{2} + \lim_{x \to d} \frac{f(x)f'''(x)}{2f'(x)f''(x)}$$

L'hopital again:

$$= \frac{1}{2} + \lim_{x \to d} \frac{f'(x)f'''(x) + f(x)f''''(x)}{2(f''(x))^2 + 2f'''(x)f'(x)}$$

$$= \frac{1}{2} + \frac{0 + 0}{2f''(d)^2} = \frac{1}{2}$$

So $|x_{i+1} - d| \leq \frac{1}{2}|x_i - d| + \frac{1}{2}|x_i - d|^2 g''(\xi)$

$$\Rightarrow \frac{|x_{i+1} - d|}{|x_i - d|} \leq \frac{1}{2} + |x_i - d| g''(\xi)$$

$\longrightarrow \frac{1}{2}$ So linearly convergent.

If $f''(\alpha) = 0$ also, then it will still be linearly convergent as the $\frac{1}{2}$ terms from above will survive.

d)

$$|x_{i+1} - \alpha| \leq |x_i - \alpha| \, g'(\alpha) + \frac{1}{2} |x_i - \alpha|^2 \max_{\xi \in (x_i, \alpha)} |g''(\xi)|$$

$$g'(\alpha) = \frac{f(x) f''(x)}{f'(x)^2} = 0$$

$$g''(\alpha) = \frac{f'(x)^2 \left(f'(x) f''(x) - f(x) f''(x) (2 f'(x) f''(x)) \right)}{f'(x)^4}$$

$$= 0$$

So we need to extend the taylor expansion:

$$|x_{i+1} - \alpha| \leq C |x_{i+1} - \alpha|^3 + |x_{i+1} - \alpha|^3 (\ldots)$$

$$\Rightarrow \frac{|x_{i+1} - \alpha|}{|x_{i+1} - \alpha|^3} \to L$$ so it converges with cubic order

3. Apply Newton's method with $x_0 = 1/2$ to find the root $\alpha = 0$ of the following functions:

(a) $f(x) = \sin x \sinh x$

(b) $f(x) = x - \sin x$

(c) $f(x) = x \cos x$

Terminate the iteration when $|f(x)| < 10^{-8}$. In each case, relate the order of convergence observed to the previous question.

$f'(x) = \sin x \cosh x + \cos x \sinh x$

$f''(x) = \cos x \cosh x + \sin x \sinh x$
$+ \cos x \cosh x - \sin x \sinh x$

$f'(x) = 0$, $f''(x) = 0$

Q2 predicts linear convergence

b) $f'(x) = 1 - \cos x$

$f''(x) = \sin x$

$\Rightarrow f'(x) = 0$

$f''(x) = 0$

$f'''(x) = 1$

So Q2 predicts linear convergence

c) $f(x) = x\cos x$

$f'(x) = -x\sin x + \cos x$

$f''(x) = -\sin x - x\cos x - \sin x$

$\Rightarrow f'(0) = 1, \quad f''(0) = 0$

So we expect cubic OOC.

5. Consider the minimisation of the energy $E : \mathbb{R}^2 \to \mathbb{R}$

$$E(x, y) = (1 - x)^2 + 100(y - x^2)^2.$$

(a) By inspection identify the global minimum.

(b) Apply your code for Newton's method from the previous question to find a root of the gradient ∇E. How many iterations does Newton's method take starting from initial guesses $(x_0, y_0) = (2, 2)$? $(10, 10)$? $(100, 100)$? Terminate the iteration when $\sqrt{\nabla E(x) \cdot \nabla E(x)} < 10^{-8}$.

(c) Using the Jacobian approximation $J(x) = \alpha I$ for some step size $\alpha > 0$ implements the gradient descent algorithm. How many iterations does gradient descent take to reach the same tolerance as part (b), starting from intial guess $(x_0, y_0) = (2, 2)$, and $\alpha = 1000$? What happens for $\alpha = 500$, and $\alpha = 2000$?

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